

MONTGOMERY, AL, USA

WMO: 722260

Lat: 32.300N

Lon: 86.408W

Elev: 62

StdP: 100.59

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.3	-2.3	-12.5	1.3	0.2	-9.9	1.6	1.7	10.1	9.6	9.0	10.2	2.4	310	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	35.9	24.5	34.8	24.4	33.8	24.3	26.5	32.7	25.8	31.8	25.5	31.3	3.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.0	28.9	24.3	19.4	28.5	23.9	18.9	28.0	82.4	32.7	80.1	32.1	78.0	31.3	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.4	6.3	DB	-8.0	37.2	2.1	1.5	-9.6	38.3	-10.8	39.2	-12.0	40.1	-13.5	41.1	(4)
(5)			WB	-8.9	27.6	2.1	1.2	-10.4	28.4	-11.6	29.1	-12.8	29.8	-14.3	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.0	8.7	11.2	14.9	18.7	23.0	26.6	27.9	27.8	25.2	19.5	13.6	10.1
	DBStd	7.93	5.67	5.32	4.81	3.69	3.05	2.04	1.54	1.77	2.82	4.30	4.71	5.23
	HDD10.0	238	94	45	14	1	0	0	0	0	0	1	18	66
	HDD18.3	1133	301	205	126	41	5	0	0	0	1	40	155	259
	CDD10.0	3512	54	79	166	261	402	497	555	553	457	294	125	68
	CDD18.3	1365	2	6	19	51	149	247	297	294	208	75	12	4
	CDH23.3	13325	9	32	170	486	1421	2487	3072	3006	1893	644	88	18
	CDH26.7	5899	0	2	25	113	550	1184	1499	1477	841	194	12	1
Wind	WSAvg	2.6	3.1	3.2	3.2	2.8	2.4	2.2	2.0	2.0	2.4	2.4	2.5	2.9
Precipitation	PrecAvg	1323	118	131	147	106	98	101	129	93	101	65	100	129
	PrecMax	1944	284	340	315	397	324	367	254	265	240	192	259	359
	PrecMin	901	28	28	22	13	6	8	33	18	16	0	13	26
	PrecStd	250	55	67	74	76	65	68	55	47	56	47	57	66
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	26.6	29.2	30.7	34.0	36.4	36.9	37.8	36.1	32.4	28.3	25.6
		MCWB	18.0	19.2	19.0	20.5	23.0	24.1	24.7	24.0	23.6	22.2	20.3	19.4
	2%	DB	22.2	24.1	27.0	29.4	32.5	35.1	35.6	36.0	34.4	30.5	25.7	23.2
		MCWB	17.2	17.9	18.3	20.2	22.5	24.0	25.0	24.6	23.3	21.3	18.9	18.6
	5%	DB	20.1	22.0	25.3	28.0	31.2	33.9	34.4	34.6	32.9	28.9	23.9	21.1
		MCWB	16.2	16.4	17.5	19.6	22.1	23.9	24.9	24.7	23.3	20.7	17.9	17.4
	10%	DB	17.9	20.1	23.3	26.4	29.9	32.5	33.2	33.3	31.4	27.1	22.0	18.9
		MCWB	14.8	15.8	16.5	18.6	21.5	23.7	24.7	24.6	23.0	19.8	17.1	15.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.9	20.8	21.2	22.9	25.1	26.6	27.3	27.3	25.9	24.6	22.6	21.3
		MCDB	22.4	24.7	25.2	28.2	30.6	32.6	33.5	33.6	31.4	28.9	25.5	23.8
	2%	WB	18.5	19.4	20.1	21.9	24.1	25.8	26.5	26.4	25.2	23.6	20.9	19.5
		MCDB	21.2	22.5	24.3	26.8	29.5	31.9	32.8	32.6	30.7	27.8	23.4	21.9
	5%	WB	17.0	18.2	19.0	21.0	23.4	25.2	25.9	25.8	24.6	22.6	19.6	18.1
		MCDB	19.4	21.2	23.5	25.5	28.7	31.1	32.0	31.7	29.8	26.3	22.8	20.4
	10%	WB	15.4	16.7	17.8	20.1	22.7	24.7	25.5	25.3	24.1	21.5	18.0	16.5
		MCDB	17.7	19.3	22.0	24.5	27.9	30.2	31.3	31.1	28.9	25.0	20.9	18.7
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.1	12.7	13.0	12.3	11.2	10.7	10.9	11.4	12.8	13.3	11.5
		MCDBR	13.2	13.4	14.4	14.2	13.4	13.1	12.2	12.6	13.0	13.8	14.3	13.1
		MCWBR	8.8	8.0	7.1	6.4	5.0	4.2	3.9	3.9	4.3	6.0	8.1	8.8
	5% WB	MCDBR	11.0	11.3	11.7	11.3	11.4	11.1	10.7	11.0	10.6	10.2	11.3	10.9
		MCWBR	8.4	8.1	6.9	6.1	5.0	4.4	4.0	4.1	4.2	5.4	7.5	8.4
Clear-Sky Solar Irradiance	taub	0.324	0.331	0.360	0.390	0.440	0.478	0.536	0.525	0.433	0.374	0.347	0.326	
	taud	2.521	2.504	2.428	2.336	2.244	2.187	2.032	2.059	2.301	2.448	2.464	2.526	
	Ebn at Noon	900	927	919	900	854	817	768	771	834	867	862	876	
	Edh at Noon	85	95	111	126	140	148	172	165	124	100	90	81	
All-Sky Solar Radiation	RadAvg	2.80	3.37	4.53	5.66	6.22	6.16	5.98	5.55	4.93	4.19	3.17	2.40	
	RadStd	0.20	0.41	0.41	0.41	0.44	0.46	0.41	0.32	0.41	0.52	0.30	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.55	N/A	N/A	N/A	N/A	N/A	-46	-101	+196	+137
Regional (40 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	+96

Nomenclature: See separate page